

Password Security Demo

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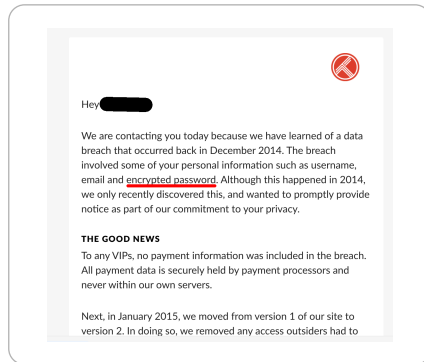
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- **Password hygiene/good practices** include among other things:
 - Passwords should be **long**, i.e., made of numerous characters/words **AND**
 - Passwords should be **complex**, i.e., made of various types of characters

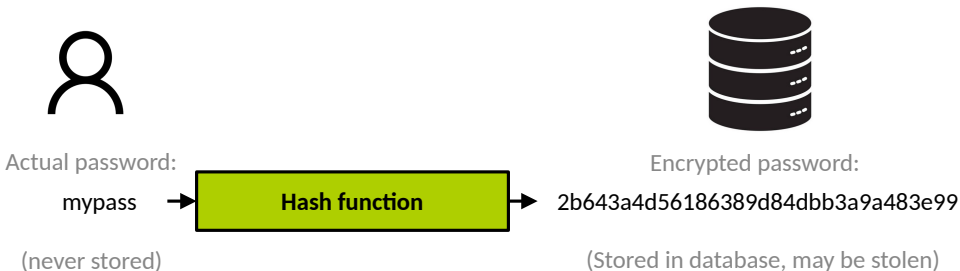
Why such rules?

Data Breaches

- Data breaches are very common
- One of the worst type of personal data that can be stolen are **passwords**

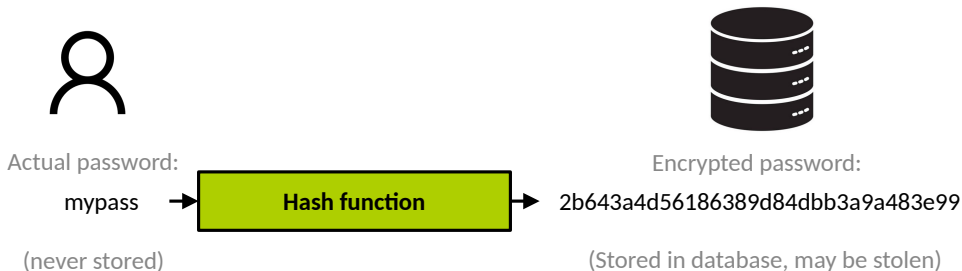


What's an Encrypted Password?



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- Authentication process:
 1. Prompt user for plain text password attempt
 2. Hash the attempt
 3. Compare to the hash of the correct password in the database: if they match, authentication succeeds

Cracking Passwords



Steal
hashes

5d41402abc4b2a76b9719d911017c592
76a2173be6393254e72ffa4d6df1030a
5ebe2294ecd0e0f08eab7690d2a6ee69
a029d0df84eb5549c641e04a9ef389e5
84675f2baf7140037b8f5afe54eef841
098f6bcd4621d373cade4e832627b4f6
385aa5385e83efc5d83c7588ee5fb7e8
4f09daa9d95bcb166a302407a0e0babe
49f68a5c8493ec2c0bf489821c21fc3b
1a1dc91c907325c69271ddf0c944bc72



Crack
passwords

hello
passwd
secret
mypass
pierre
test
phrase
short
hi
pass



- Hackers commonly steal lists of **hashed passwords** from online services and applications
- Their next goal is then to **revert the hashes into plain text passwords**

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1. **Bruteforce attack:** try every possible combination of characters
 2. **Dictionary attack:** try commonly used passwords, e.g., dictionary words

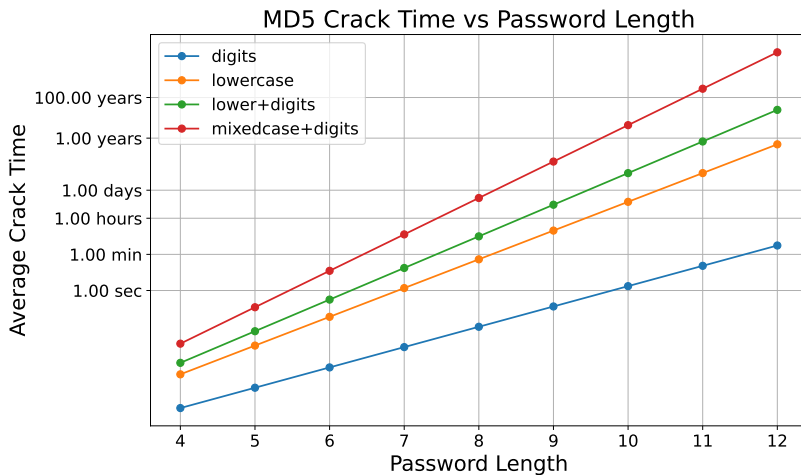
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- Very efficient on weak passwords with the processing power of modern computers

Demo 1: Cracking a Simple Password

MD5 Cracking Speed



Demo 2: Cracking a List of Hashes

Good practices for strong password security **make it very hard/impossible to revert hashes**

- On the end user side:
 - **Choose long (12+ characters) passwords**
 - Choose passwords with a wide set of characters: digits, special characters, etc.
- On the application/website developer side:
 - Force user to choose long/varied passwords
 - Use strong hash methods, make it at least a bit computationally expensive to hash
 - Combine passwords with random data (salt) before hashing